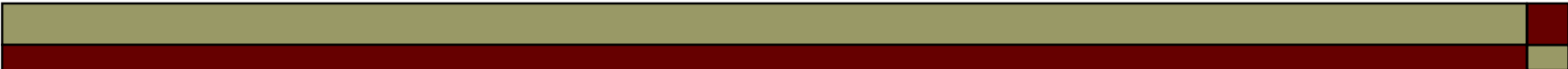


Software Construction and Development SE-312

LECTURE # 02

Software process Improvement

Course Instructor: Engr. Ruqqiya Asad



Objectives

- ❑ To explain the principles of software process improvement.
- ❑ To explain how software process factor influence software quality and productivity.
- ❑ To explain how to develop simple model of software processes.
- ❑ To explain the notion of process capability and the CMMI process improvement model.

Information Systems Development



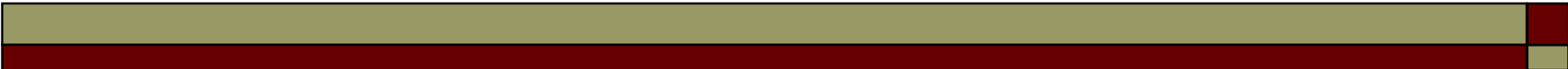
Resources → *Activities* → *Products*



equipment

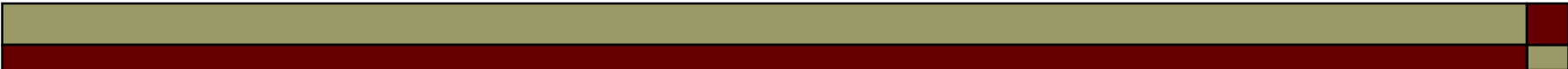
- Planning
- Analysis
- Design
- Construction
- Testing
- Training
- Implementation
- Follow-up
- Enhancements
- etc...

- Hardware
- Software
- Documentation



System Development Process

A system development process is the set of activities, methods, , practices , transformations and methods that developers used to develop and maintain information systems.



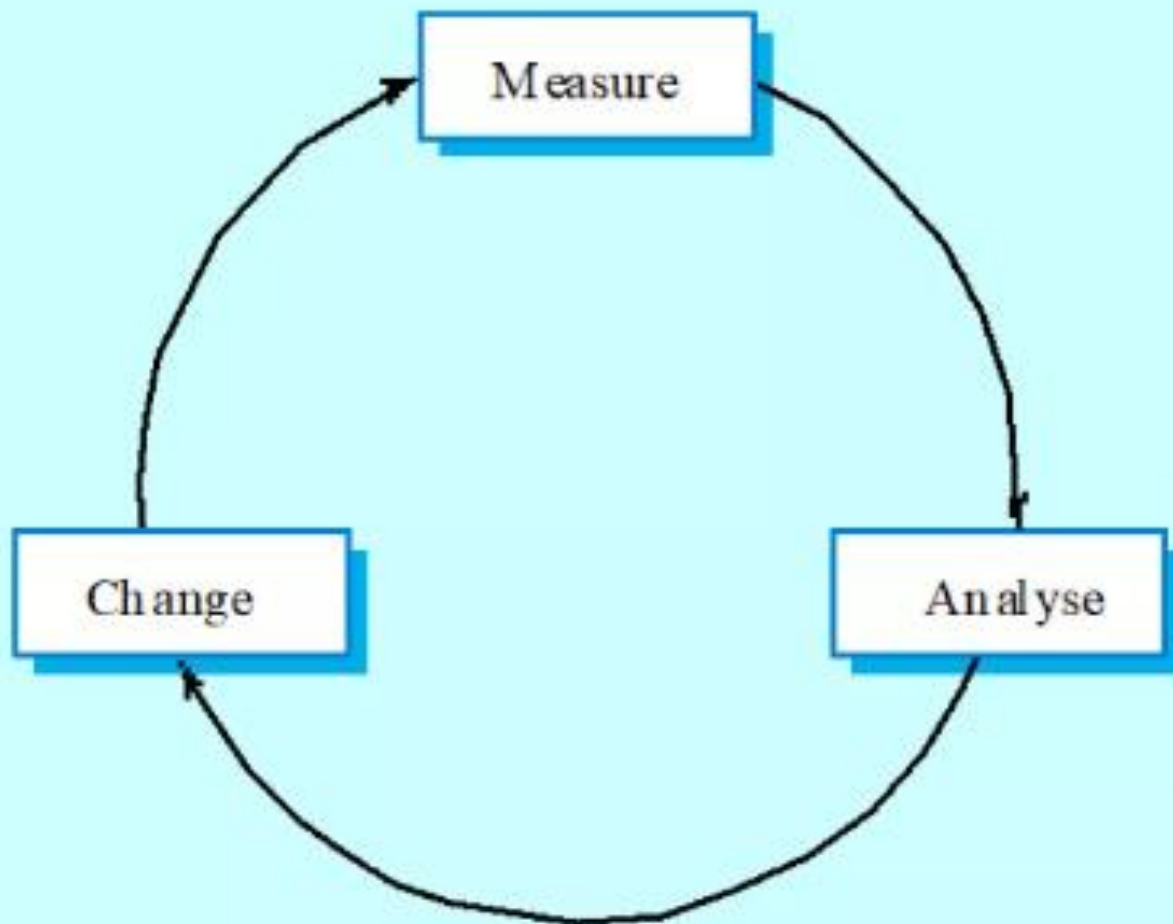
Software Process Improvement

Software process improvement is the name given to the identification of the current state-of-the-practice of information systems development within an organization and then improving it.

Process Improvement Approach



The process improvement cycle



Process improvement stages

- **Process measurement**
 - Attributes of the current process are measured. These are a baseline for assessing improvements.
- **Process analysis**
 - The current process is assessed and bottlenecks and weaknesses are identified.
- **Process change**
 - Changes to the process that have been identified during the analysis are introduced.

Examples: -



Some example results of an improved software development process could include:

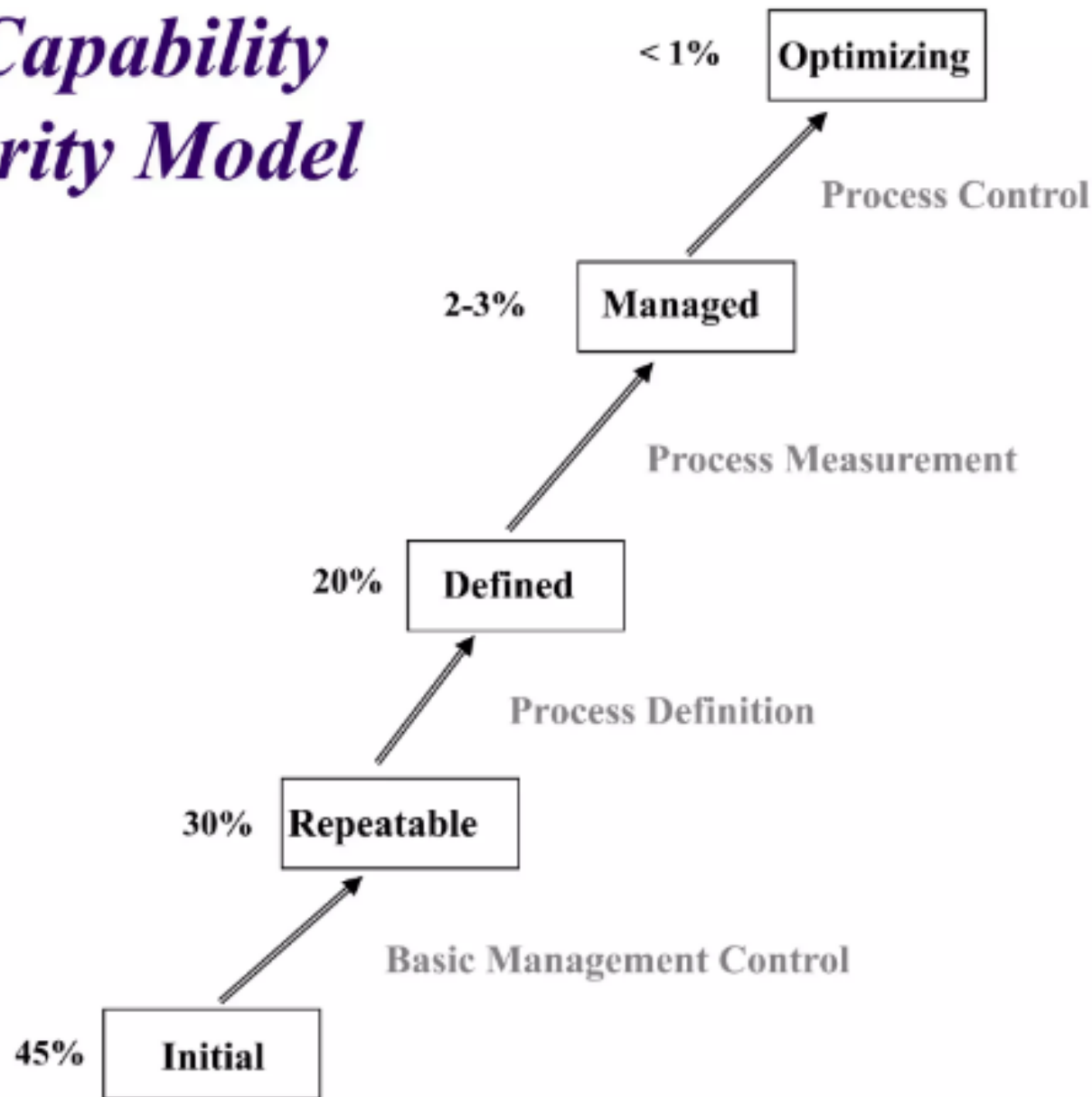
- Fewer product defects found by customers.
- Earlier identification and correction of defects.
- Fewer defects Introduced during the development process.
- Faster time to market.
- Better predictability of project schedules and resources.

Software Process Improvement Efforts



- *Carnegie Mellon University's Software Engineering Institute's Capability Maturity Model - (SEI's CMM)*
- *International Standards Organization's 9001 Specification (ISO 9001)*
- *Proprietary SPI's from consulting firms*

SEI Capability Maturity Model



CMM - Initial (Level 1)



- *The software process is characterized as ad hoc, occasionally even chaotic*
- *Few processes are defined*
- *Success depends on individual effort and heroics*

“BASICALLY NO CONTROL”

CMM - Repeatable (Level 2)

- *Basic project management processes are established to track cost, schedule, and functionality*
- *The necessary process discipline is in place to repeat earlier successes on projects with similar applications*
- *Success achieved through basic project management; not advanced technologies*

“BASIC MANAGEMENT CONTROL”

CMM - Defined (Level 3)

- *The software process for both management and engineering activities is documented, standardized, and integrated into a standard software process for the organization*
- *All projects use an approved, tailored version of the organization's standard software process for developing and maintaining software*
- *Formality lends itself to improvement*

“PROCESS DEFINITION”

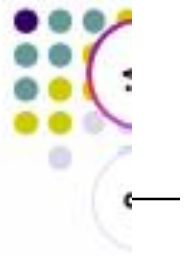
CMM - Managed (Level 4)



- *Detailed measures of the software process and product quality are collected*
- *Both the software process and products are quantitatively understood and controlled*
- *A software metrics program is in use*

“PROCESS MEASUREMENT”

CMM - Optimizing (Level 5)



- *Continuous process improvement is enabled by quantitative (metrics) feedback from the process*
- *Continuous process improvement is enabled by piloting innovative ideas and technologies*

“PROCESS CONTROL”